# House Price Prediction Using Machine Learning

## Student

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## Project Overview

This project predicts house prices using a Machine Learning model trained on the California Housing Dataset. A Streamlit web application allows users to enter housing features and receive an estimated house price instantly.

## Objective

To build a machine learning application that estimates house prices accurately using historical housing data.

## Technologies Used

Python, Streamlit, Pandas, NumPy, Scikit-learn, Matplotlib.

## Dataset

California Housing Dataset (Scikit-learn). Features: Median Income, House Age, Average Rooms, Average Bedrooms, Population, Average Occupancy, Latitude and Longitude. Target: Median House Value.

## Methodology

Load dataset.
Preprocess with MinMaxScaler.
Train Linear Regression model.
Save trained model (.pkl).
Predict using Streamlit.

## Results

The application predicts estimated house prices from user-provided housing features using the trained model.

## Future Scope

Improve accuracy, compare multiple ML models, deploy online, and add visualization dashboards.

## Conclusion

This project demonstrates the use of Machine Learning and Streamlit to solve a real-world house price prediction problem.